

DD_FA-01 - Equipment Field Audit - Developer Implementation Guide

Version: v1.0 Bundle: 10_other Companion DD: DD_FA-01-Equipment_Field_Audit-v1.0

1. Goal

Implement the standalone equipment field audit process.

The implementation must let operations create audit tasks, freeze expected equipment from OSR-INSTANCE, let field users capture observations from the mobile app, compare expected vs observed state, and route discrepancies to the correct module.

Do not use FA-01 as a replacement for every normal work order. Installation, shifting, support, and swap flows keep their own WO evidence. FA-01 is for planned audits and exception-driven verification.

2. Runtime Flow

```
sequenceDiagram
    participant BO as Backoffice Audit Console
    participant FA as FA-01
    participant INST as OSR-INSTANCE
    participant WO as WO-01
    participant MOB as Contractor Mobile App
    participant RULES as Drools
    participant TCK as TCK-01
    participant RMA as OSR-RMA
    participant APR as EM-CFG-04

    BO->>FA: POST /api/equipment-field-audits/tasks
    FA->>INST: GET expected equipment by customer/subscription/homepass
    FA->>FA: create expected item snapshot
    FA->>WO: optional POST /api/work-orders kind FIELD_AUDIT
    MOB->>FA: GET assigned tasks
    MOB->>FA: POST observations with serial/photo/GPS
    MOB->>FA: POST submit
    FA->>FA: compare expected vs observed
    FA->>RULES: severity/routing decision
    FA->>TCK: create ticket when service/customer issue
    FA->>RMA: create recovery/RMA when equipment action needed
    FA->>APR: approval for write-off/state correction
```

3. Step 1 - Database Migrations

Create:

- equipment_field_audit_campaign
- equipment_field_audit_task
- equipment_field_audit_expected_item
- equipment_field_audit_observation
- equipment_field_audit_discrepancy

Implementation details:

- Add a unique index for mobile offline submissions: (operator_code, offline_client_ref) on observations.
- Store expected state snapshots as JSON so the audit remains explainable even if OSR-INSTANCE changes later.
- Store only file ids in observations, not binary photo content.
- Add indexes for assigned user task lists and discrepancy work queues.

4. Step 2 - Create Campaign

Backoffice calls:

```
POST /api/equipment-field-audits/campaigns
Content-Type: application/json

{
  "operatorCode": "WIK",
  "campaignType": "CUSTOMER_EQUIPMENT_VERIFY",
  "areaCode": "NRB-WEST",
  "technologyFamily": "GPON",
  "scheduledStartAt": "2026-08-10T08:00:00Z",
  "scheduledEndAt": "2026-08-20T17:00:00Z"
}
```

FA-01:

1. Validates permissions through EM-CFG-03.
2. Creates `equipment_field_audit_campaign` in SCHEDULED.
3. Emits `EquipmentFieldAuditCampaignCreated`.

Campaigns are optional. A support user may create one direct audit task without a campaign.

5. Step 3 - Create Audit Task And Freeze Expected Items

Backoffice or another module calls:

```
POST /api/equipment-field-audits/tasks
Idempotency-Key: efat-CUS-100045-202608
Content-Type: application/json

{
  "operatorCode": "WIK",
  "campaignId": "efac-001",
  "taskType": "CUSTOMER_PREMISES",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "homepassId": "HP-GPON-0440",
  "assignedToUserId": "usr-tech-010",
  "dueAt": "2026-08-12T17:00:00Z",
  "createWorkOrder": true
}
```

FA-01:

1. Validates request and idempotency key.
2. Creates `equipment_field_audit_task` in CREATED.
3. Calls OSR-INSTANCE to load expected equipment:

```
GET /api/equipment-instances?operatorCode=WIK&customerId=CUS-100045&subscriptionId=SUB-700045&state=INSTALLED
```
4. For each returned equipment instance, inserts `equipment_field_audit_expected_item`.
5. If no expected item exists, task can still be created for investigation, but mark it with a warning.
6. Emits `EquipmentFieldAuditTaskCreated`.

Expected item sample from OSR-INSTANCE:

```
{
  "equipmentInstanceId": "inst-gpon-001",
  "skuId": "WIK-ONT-HUAWEI-EG8145V5",
  "serialNumber": "ONT123456",
  "state": "INSTALLED",
  "binding": {
    "customerId": "CUS-100045",
    "subscriptionId": "SUB-700045",
    "homepassId": "HP-GPON-0440"
  }
}
```

6. Step 4 - Create Work Order When Dispatch Is Needed

If createWorkOrder=true, call WO-01:

```
POST /api/work-orders
Idempotency-Key: efat-001-wo
Content-Type: application/json

{
  "operatorCode": "WIK",
  "kind": "FIELD_AUDIT",
  "originatingContextType": "EQUIPMENT_FIELD_AUDIT_TASK",
  "originatingContextId": "efat-001",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "homepassId": "HP-GPON-0440",
  "metadata": {
    "requiredEvidence": ["serialScan", "photos", "gps", "condition"],
    "expectedSerials": ["ONT123456"]
  }
}
```

FA-01 stores wo_id and moves the task to ASSIGNED.

If the operator does not want a WO for simple audits, assign the task directly to a user/team and expose it through the mobile task API.

7. Step 5 - Mobile App Downloads Task

Mobile app calls:

```
GET /api/equipment-field-audits/tasks?assignedToUserId=usr-tech-010&status=ASSIGNED
```

Response:

```
{
  "items": [
    {
      "auditTaskId": "efat-001",
      "customerId": "CUS-100045",
      "subscriptionId": "SUB-700045",
      "homepassId": "HP-GPON-0440",
      "dueAt": "2026-08-12T17:00:00Z",
      "expectedItems": [
        {
          "expectedItemId": "efae-001",
          "skuId": "WIK-ONT-HUAWEI-EG8145V5",
          "serialNumber": "ONT123456",
          "expectedCondition": "INSTALLED_WORKING"
        }
      ]
    }
  ]
}
```

Mobile app must support:

- serial/QR scan;
- manual serial entry with reason;
- photo upload through foundation file storage;
- GPS capture;
- offline client reference for retry-safe submission.

8. Step 6 - Submit Observations

Mobile app uploads files first and receives file ids. Then it calls:

```
POST /api/equipment-field-audits/tasks/efat-001/observations
Idempotency-Key: mob-obs-7788
Content-Type: application/json

{
```

```

    "expectedItemId": "efae-001",
    "observedSerialNumber": "ONT123456",
    "observedSkuId": "WIK-ONT-HUAWEI-EG8145V5",
    "presenceStatus": "PRESENT",
    "conditionStatus": "WORKING",
    "photoFileIds": ["file-001", "file-002"],
    "gpsLatitude": -1.2921,
    "gpsLongitude": 36.8219,
    "notes": "ONT installed beside router.",
    "capturedAt": "2026-08-12T10:15:00Z",
    "offlineClientRef": "mob-obs-7788"
}

```

FA-01:

1. Checks task assignment and status.
2. Checks idempotency by offline client ref.
3. Validates file ids exist and are attached to the current operator.
4. Stores equipment_field_audit_observation.
5. Does not compare or route until task submit, unless the operator wants immediate discrepancy preview.

9. Step 7 - Submit Task And Compare

Mobile app calls:

```

POST /api/equipment-field-audits/tasks/efat-001/submit
Content-Type: application/json

{
  "submittedByUserId": "usr-tech-010",
  "submittedAt": "2026-08-12T10:30:00Z"
}

```

FA-01 comparison logic:

1. Load expected items.
2. Load observations.
3. For each expected item:
 - no matching observation means MISSING;
 - different serial means WRONG_SERIAL;
 - conditionStatus=DAMAGED means DAMAGED;
 - GPS too far from expected location may mean WRONG_LOCATION.
4. For each observation without expected match, create FOUND_EXTRA.
5. If no discrepancy, set task CLOSED and emit EquipmentFieldAuditTaskClosed.
6. If discrepancies exist, set task DISCREPANCY_OPEN and emit EquipmentFieldAuditDiscrepancyOpened.

Do not change OSR-INSTANCE state during comparison.

10. Step 8 - Route Discrepancies

Run Drools for every discrepancy:

```

public record EquipmentFieldAuditFact(
    String operatorCode,
    String discrepancyType,
    String expectedLocationType,
    String observedCondition,
    boolean equipmentRecoverable,
    boolean customerServiceImpacted,
    boolean serialBelongsToAnotherCustomer,
    String technologyFamily
) {}

```

Severity rules:

```

package rules.field_audit.equipment.severity

rule "Wrong serial assigned to another customer is critical"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "WRONG_SERIAL",
                                  serialBelongsToAnotherCustomer == true)
    $d : EquipmentFieldAuditDecision()
then
    $d.setSeverity("CRITICAL");
end

rule "Missing customer premises equipment is high severity"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "MISSING",
                                  expectedLocationType == "CUSTOMER_PREMISES")
    $d : EquipmentFieldAuditDecision()
then
    $d.setSeverity("HIGH");
end

```

Routing rules:

```

package rules.field_audit.equipment.routing

rule "Damaged equipment with customer impact creates ticket"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "DAMAGED",
                                  customerServiceImpacted == true)
    $d : EquipmentFieldAuditDecision()
then
    $d.route("CREATE_TICKET");
end

rule "Recoverable missing equipment creates RMA recovery"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "MISSING",
                                  equipmentRecoverable == true)
    $d : EquipmentFieldAuditDecision()
then
    $d.route("CREATE_RMA_RECOVERY");
end

rule "Unrecoverable equipment requests write off approval"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "MISSING",
                                  equipmentRecoverable == false)
    $d : EquipmentFieldAuditDecision()
then
    $d.route("REQUEST_WRITE_OFF");
    $d.requireApproval("EQUIPMENT_FIELD_AUDIT_WRITE_OFF");
end

```

11. Step 9 - Execute Route Actions

11.1 Create Ticket

Use when the discrepancy is customer/service-facing:

```

POST /api/tickets
Idempotency-Key: efad-001-ticket
Content-Type: application/json

{
  "operatorCode": "WIK",
  "ticketType": "EQUIPMENT_FIELD_AUDIT_ISSUE",
  "customerId": "CUS-100045",
  "subscriptionId": "SUB-700045",
  "originRefType": "EQUIPMENT_FIELD_AUDIT_DISCREPANCY",
  "originRefId": "efad-001",
  "summary": "Damaged ONT found during field audit"
}

```

11.2 Create Recovery Or RMA

Use when equipment must be recovered, swapped, or investigated physically:

```

POST /api/osr-rma/requests
Idempotency-Key: efad-001-rma
Content-Type: application/json

```

```
{
  "operatorCode": "WIK",
  "requestType": "EQUIPMENT_RECOVERY",
  "originRefType": "EQUIPMENT_FIELD_AUDIT_DISCREPANCY",
  "originRefId": "efad-001",
  "equipmentInstanceId": "inst-gpon-001",
  "customerId": "CUS-100045",
  "reasonCode": "FIELD_AUDIT_MISSING"
}
```

11.3 Request OSR-INSTANCE Correction

Use only when evidence proves OSR-INSTANCE state is wrong:

```
POST /api/equipment-instances/inst-gpon-001/lifecycle-corrections
Idempotency-Key: efad-001-correction
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "originRefType": "EQUIPMENT_FIELD_AUDIT_DISCREPANCY",
  "originRefId": "efad-001",
  "correctionType": "LOCATION_CORRECTION",
  "targetLocationType": "CUSTOMER_PREMISES",
  "targetLocationRef": "CUS-100045",
  "evidenceFileIds": ["file-001", "file-002"]
}
```

If rules or policy require approval, call EM-CFG-04 before this correction.

11.4 Request Write-Off Approval

```
POST /api/approval-requests
Idempotency-Key: efad-001-writeoff-approval
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "approvalCode": "EQUIPMENT_FIELD_AUDIT_WRITE_OFF",
  "originRefType": "EQUIPMENT_FIELD_AUDIT_DISCREPANCY",
  "originRefId": "efad-001",
  "requestedByUserId": "usr-audit-manager",
  "payload": {
    "equipmentInstanceId": "inst-gpon-001",
    "serialNumber": "ONT123456",
    "reason": "Missing after field audit and marked unrecoverable"
  }
}
```

After approval, FA-01 routes the actual write-off to OSR-05/OSR-INSTANCE according to the existing inventory governance. FA-01 does not write off equipment directly.

12. Step 10 - Events

Emit through outbox:

- EquipmentFieldAuditCampaignCreated
- EquipmentFieldAuditTaskCreated
- EquipmentFieldAuditTaskAssigned
- EquipmentFieldAuditSubmitted
- EquipmentFieldAuditDiscrepancyOpened
- EquipmentFieldAuditDiscrepancyRouted
- EquipmentFieldAuditTaskClosed

Each event must include:

- operatorCode
- auditTaskId or campaignId
- customerId and subscriptionId when available

- correlationId
- causationId

13. Step 11 - Tests

Build tests in this order:

1. Create task and freeze expected items from OSR-INSTANCE.
2. Same idempotency key returns same task.
3. Mobile observation duplicate offline ref returns same observation.
4. Matching observation closes the task.
5. Missing expected item creates MISSING discrepancy.
6. Extra serial creates FOUND_EXTRA discrepancy.
7. Wrong serial belonging to another customer becomes CRITICAL.
8. Damaged equipment with service impact creates a ticket.
9. Missing recoverable equipment creates OSR-RMA recovery.
10. Write-off route calls EM-CFG-04 and waits for approval.
11. FA-01 never calls OSR-02 procurement.

14. Operational Notes

- Mobile app uploads may arrive after the field visit because connectivity can be poor. Use offline refs everywhere.
- Keep evidence immutable after submission. If a reviewer adds notes, create review comments instead of changing field evidence.
- Do not let field users directly mark equipment written off or corrected.
- Keep discrepancy routing visible in backoffice so audit managers can see the downstream ticket/RMA/approval reference.
- Reporting should use FA-01 events and discrepancy tables, not screenshots from the mobile app.